

# Introduction to Machine Learning

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# Recommended Textbook

- An Introduction to Statistical Learning with Application in R (or in Python)
  - [Book website](#)
  - [Online course videos](#)
  - [Notes from a book club from the R4DS online learning community](#)
- Elements of Statistical Learning
  - [Book website](#)
  - Recommended for advanced students

- **R** – we will use R mostly in this class  
Download [R](#), and install  
Download [RStudio](#), and install
- **Python** – strongly recommended
  - Download and install [Anaconda](#)
  - [Get started with Anaconda](#)
  - Jupyter Notebooks is recommended environment (like RStudio)

# Learning Resources

- Data
  - Most commonly used public data sets
  - Textbook data (James, et al.): install R package [ISLR](#)
  - UCI Machine Learning Repository
  - Kaggle
  - KDD Nuggets
- Lectures and other tutorials
  - Videos of textbook (ISLR)
  - DataCamp
  - Coursera
- Your AI friends
  - ChatGPT, Claude AI
  - many more...
- Course website
  - a publicly available site
  - Canvas (for assignment and other work)

# You should take this course if...

- You are passionate about AI and data science
- You know that data are assets for businesses and you like exploring data
- You want to pursue this career
- You want to pursue advanced degrees in data science and AI

An [article](#) on who should pursue a master's degree in BA.

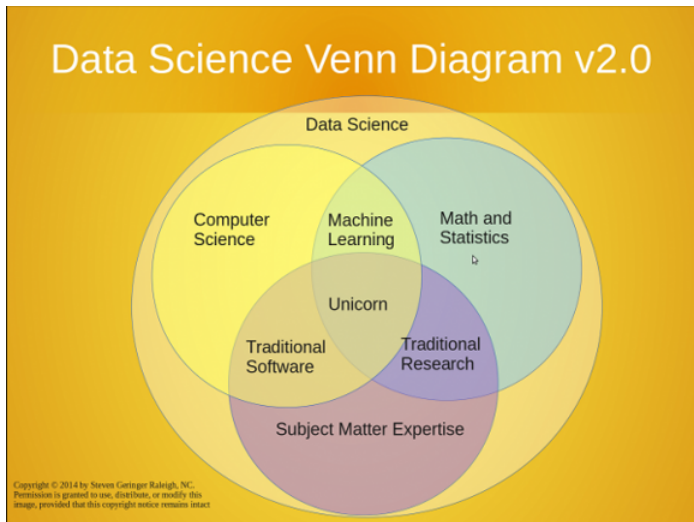
How much salary can a data scientist earn? ([a report from indeed](#))

# Machine Learning in Different Fields

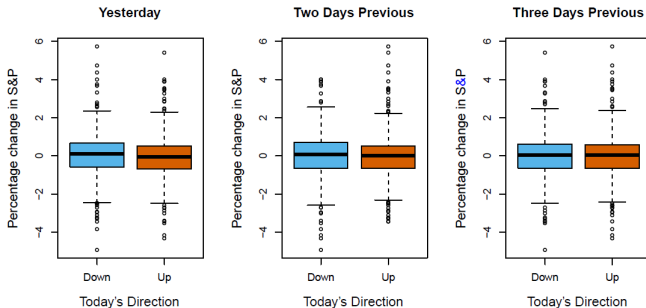
Learning from data is essential in different scientific disciplines

- Predict stock returns in next six months based on historical data;
- Predict the probability of a loan default based on customer's information and historical records;
- Identify certain diseases based on medical image;
- Identify handwritten digits from image;
- Facial recognition;
- Natural language processing;
- Cluster customers based on their purchase behavior and other information
- .....

## In practice, you need...



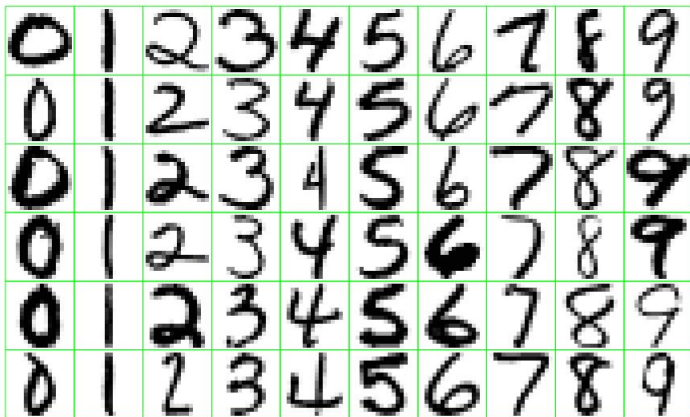
# Examples



**FIGURE 1.2.** Left: Boxplots of the previous day's percentage change in the S&P index for the days for which the market increased or decreased, obtained from the Smarket data. Center and Right: Same as left panel, but the percentage changes for 2 and 3 days previous are shown.

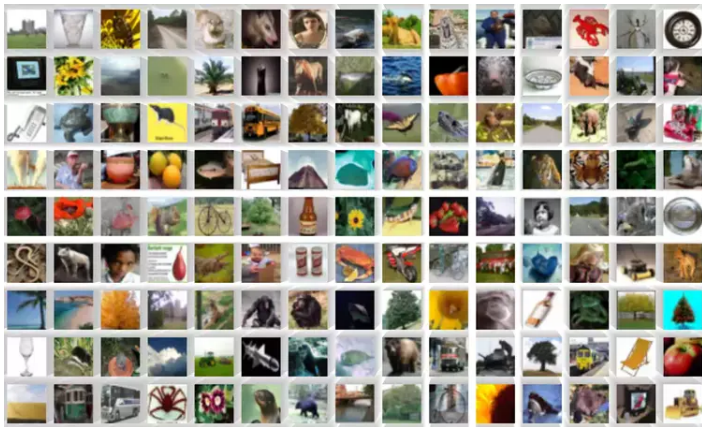
- This example is from ISLR page 3 (stock market data).
- Go to [ChatGPT](#) and ask it for a demonstrative example of analyzing the Smarket dataset from the textbook.

# Examples



- This is part of the famous [MNIST](#) dataset for image recognition.
- Go to [ChatGPT](#) and ask it for a demonstrative example of analyzing the MNIST dataset.

# Examples



- This is part of the famous [CIFAR-100](#) dataset for image recognition.
- There are 100 classes labeled manually by researchers.
- Go to [ChatGPT](#) and ask it for a demonstrative example of analyzing the CIFAR-100 dataset.

- **Supervised learning**

- There is specific response you need to predict

- **Unsupervised learning**

- No response, instead, you need to create response based on some patterns

- **Semi-supervised learning**

- Mixture of both

- **Supervised learning**
  - There is specific response you need to predict
- **Unsupervised learning**
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- **Semi-supervised learning**
  - Mixture of both
- What type of learning is for large language model?

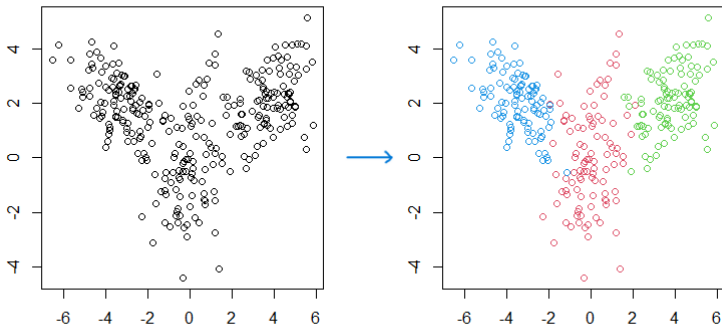
# Unsupervised Learning

- Data is **unlabeled** (no  $Y$ 's)
- Uncover patterns, groups among  $X$ 's
- Subjective, no simple goal such as prediction
- Examples:
  - Marketing segmentation – clustering
  - Dimension reduction – principle component analysis (PCA)
  - Market basket analysis – association rules mining

# A simple ML algorithm – k-means clustering

- Clustering: group similar observations together
- k-means: one out of many algorithms for clustering
  - k: how many groups
  - means: the center of the group
- This is an iterative algorithm
  - making a little progress many times
- Similarity measures, or distances, are the key component behind the algorithm

# An illustration



# Supervised Learning

- Labeled data
- The goal is to predict or explain certain outcome
- Type of problem:
  - Regression: outcome is continuous
  - Classification: outcome is categorical
- Popular ML algorithms:
  - Least square, nearest neighbor, CART, gradient boosting, neural network, deep learning

# Supervised Learning

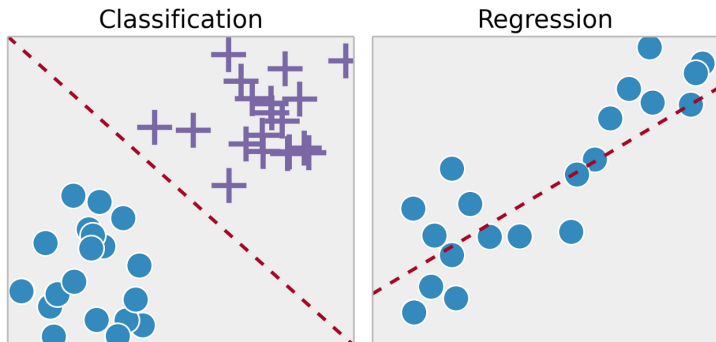
- Suppose we observe data  $Y_i$  and  $\mathbf{x}_i = (X_{i1}, \dots, X_{ip})$  for  $i = 1, \dots, n$ .
- $Y$  is the outcome (or response, dependent variable, target), and  $\mathbf{x}$  is predictor (or independent variables, covariates, features, inputs)
- The learning problem can be modeled as

$$Y_i = f(\mathbf{x}_i) + \epsilon_i$$

where  $f(\cdot)$  is unknown function, and  $\epsilon$  is random error.

- A general workflow: Learn from the **training data** and test the trained model on a holdout sample (**testing data**).

# Regression and Classification



- Response variable is continuous
  - e.g., stock return, housing price, temperature
- Goal: prediction
- Learning methods:
  - Parametric methods, e.g., linear regression
  - Nonparametric methods, e.g., decision tree, neural network
- Commonly used evaluation metrics:
  - Mean squared error (MSE), mean absolute error (MAE), R-square

# An example for regression

Can we predict Sales using ad media spending?

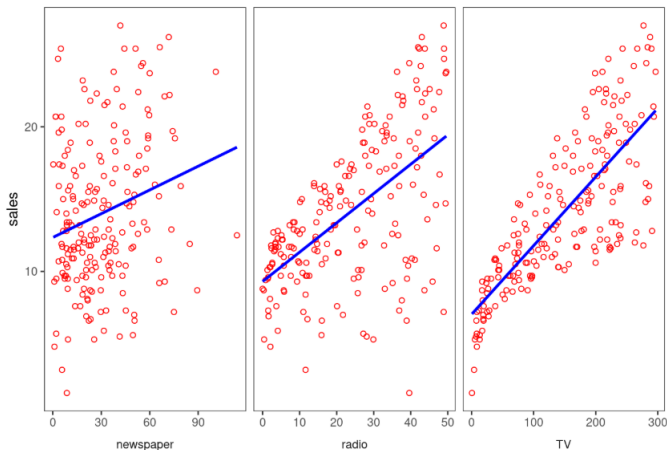
- Three channels of media: TV, radio, and newspaper
  - Spending on these channels are predictors, often denoted by  $X$
- The outcome variable is sales, often denoted by  $Y$
- Because our intuition is higher ads spending leads to higher sales. Thus, a linear regression can be developed to predict sales:

$$Y = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + \beta_3 * X_3 + \epsilon$$

- Here the unknown function is a linear function, that is
  - $f(x) = \beta_0 + \beta_1 * X_1 + \beta_2 * X_2 + \beta_3 * X_3$
- The model needs to “learn” from the data to figure out the parameters:  $\beta_0, \beta_1, \beta_2, \beta_3$ . How?

# Visualizing the data

Exploratory data analysis (EDA) is always important and a necessary step before building any models



# Assessment

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Now practice time. Let's play with this data!

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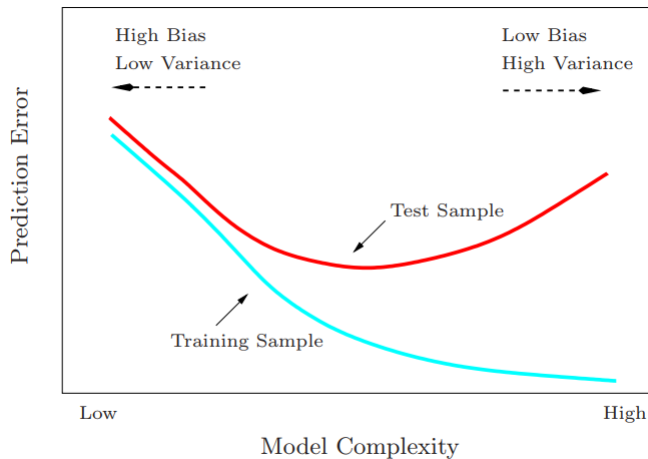
- How good is the model?
  - model fitting (in-sample performance)
  - prediction (out-of-sample performance)
- One of the most common metrics is the *mean squared error*
- Denote training set  $Tr = \{x_i, y_i\}_1^N$ , and testing set  $Te = \{x_i, y_i\}_1^M$

$$MSE_{Tr} = \frac{1}{N} \sum_{i \in Tr} (y_i - \hat{f}(x_i))^2$$

$$MSE_{Te} = \frac{1}{M} \sum_{i \in Te} (y_i - \hat{f}(x_i))^2$$

- Training error,  $MSE_{Tr}$ , may be biased due to overfitting
- In this course, we denote *MSE* as training error, and *MSPE* as testing error

# Prediction error – a tradeoff

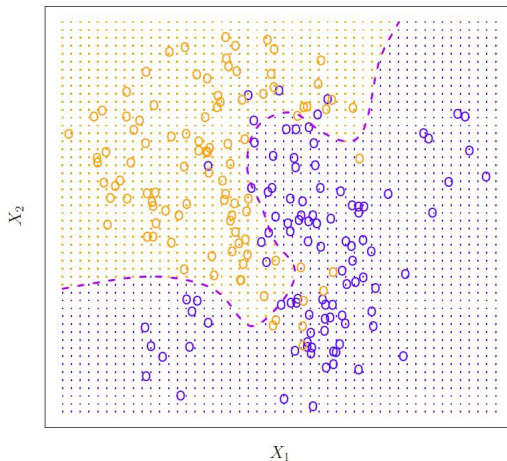




# Classification

- Response variable is categorical
  - e.g., {A, B, C}, {dog, cat}, {0, 1}
- Goal: prediction (classification)
- Learning methods:
  - Parametric methods, e.g., logistic regression
  - Nonparametric methods, e.g., decision tree, neural network
- an example of neural networks: [\[link\]](#)
- Commonly used evaluation metrics:
  - misclassification error, false positive rate (FPR), false negative rate (FNR), AUC, F1 score, pseudo R-square, etc.

# Classification – an illustration<sup>1</sup>

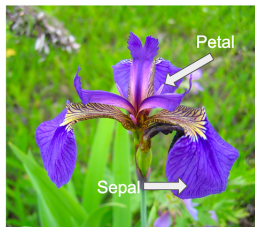


<sup>1</sup>Source: ISLR pp.38, Figure 2.13

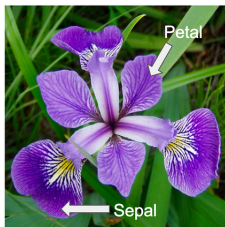
# An example for classification

Suppose we want to classify types of flowers by measuring their sizes.

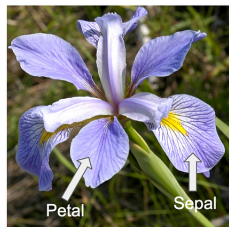
*Iris setosa*



*Iris versicolor*



*Iris virginica*



- Predictors ( $X$ ): sepal width and length, petal width and length
- Label ( $Y$ ): {setosa, versicolor, virginica}
- Learning goal: what kind of  $X$  more likely belongs to what class

# A simple classifier: K-nearest neighbor

- Model:

$$\hat{Y} = \hat{f}(\mathbf{x}) = \frac{1}{k} \sum_{\mathbf{x}_i \in N_k(\mathbf{x})} y_i$$

where  $N_k(\mathbf{x})$  is the neighborhood of  $\mathbf{x}$  defined by the  $k$  closest points  $\mathbf{x}_i$ .

- $k$ : size of the neighbor
  - smaller or bigger neighbor is better?
- How to find the neighbor of any point  $\mathbf{x}$ ?

Now practice time. Let's play with this data!

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- Same question: How good is the model?
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# Prediction accuracy for classification problems

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- Response variable is **qualitative**, which has no numeric meaning
- A classifier  $\hat{C}(x)$  either correctly predict  $x$  or not
- A commonly used prediction error: Misclassification rate (MR)

$$MR_{Te} = \frac{1}{M} \sum_{i \in Te} \mathbb{I}[y_i \neq \hat{C}(x_i)]$$

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- Most classifier  $\hat{C}(x)$  involves certain decision rule
  - As a learning outcome, the machine may say “a flower with characteristics  $x$  is more likely to be setosa than other types.”

# K-fold cross-validation

- Instead of doing train and test once, we can do it  $K$  times

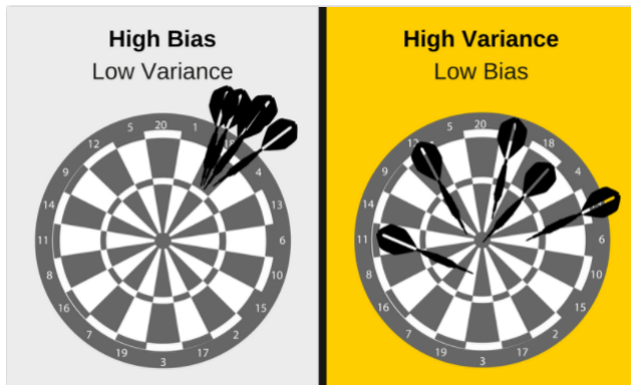
| Folder 1 | Folder 2 | Folder 3 | Folder 4 | Folder 5 |
|----------|----------|----------|----------|----------|
| Test     | Train    | Train    | Train    | Train    |
| Train    | Test     | Train    | Train    | Train    |
| Train    | Train    | Test     | Train    | Train    |
| Train    | Train    | Train    | Test     | Train    |
| Train    | Train    | Train    | Train    | Test     |

- Use 2,3,4,5 as training and 1 as testing
- Use 1,3,4,5 as training and 2 as testing
- Keep doing this loop...
- Average 5 testing errors, that is CV score
- In practice,  $K = 5$  or  $10$  is recommended

# Bias-Variance Tradeoff

- This is a very important tradeoff that governs the choice of statistical learning methods.
- **Bias:** how far the estimated model  $\hat{f}(x)$  is to the true model  $f(x)$ .
  - Unbiased estimate is defined as:  $\mathbb{E}\hat{f}(x) = f(x)$
  - Usually, we calculate the squared bias:  $(\mathbb{E}\hat{f}(x) - f(x))^2$
- **Variance:** the variation of estimated model  $\hat{f}(x)$  based on different training set.

# Bias-Variance Tradeoff



Source: [link](#)

# Bias-Variance Tradeoff

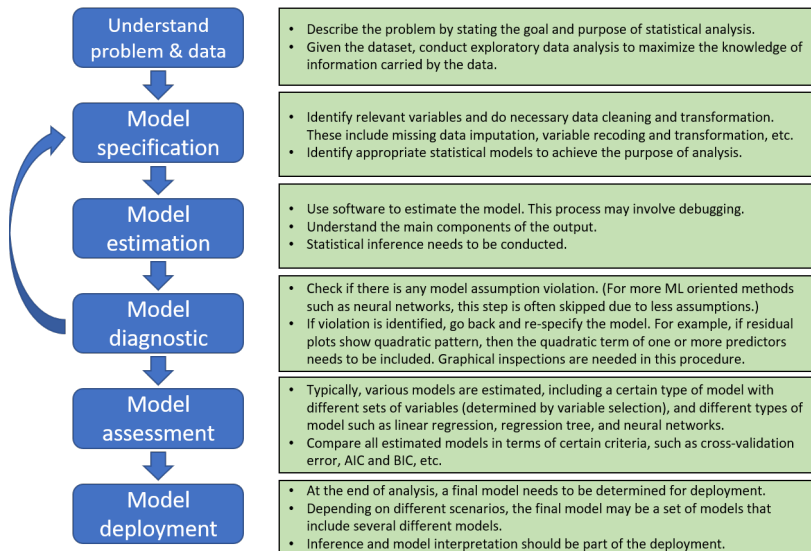
- Suppose the data arise from a model  $Y = f(x) + \epsilon$ , with  $\mathbb{E}(\epsilon) = 0$  and  $\text{Var}(\epsilon) = \sigma^2$ .
- Suppose  $\hat{f}(x)$  is trained based on some training data, and let  $(x_0, y_0)$  be a test observation from the same population.
- The *expected prediction error* can be decomposed to:

$$\mathbb{E}[y_0 - \hat{f}(x_0)]^2 = \sigma^2 + \text{Bias}^2(\hat{f}(x_0)) + \text{Var}(\hat{f}(x_0))$$

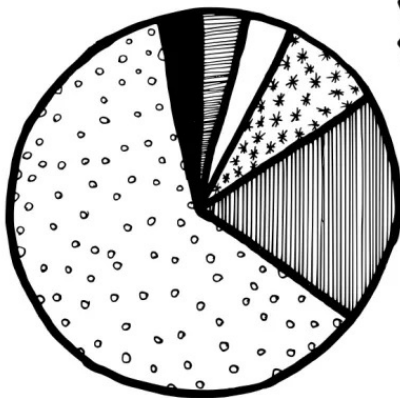
(Challenge yourself: Show it.)

- Typically as the **flexibility** of  $\hat{f}$  increases, its variance increases, and its bias decreases.

# Model building process



# The reality



## WHAT DATA SCIENTISTS SPEND THEIR TIME DOING



知乎 @李沐